



Subrata Golui

Postdoctoral Fellow

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Profile

I am a researcher in stochastic control, Markov decision processes, and stochastic games, with research interests spanning risk-sensitive control, singular perturbations, partially observable stochastic systems, and stochastic differential games. I received my M.Sc. in Applied Mathematics from the Indian Institute of Engineering Science and Technology, Shibpur, in 2016, with Operations Research as my special paper. I completed my Ph.D. in Mathematics at the Indian Institute of Technology Guwahati on July 4, 2023, under the supervision of Dr. Chandan Pal.

Currently, I am a **Postdoctoral Fellow at the Southern University of Science and Technology (SUSTech), Shenzhen, China**. In addition, I have been **invited by IIT Dharwad to serve as an instructor** for two courses, *Introduction to Mathematical Finance I* and *Stochastic Models*, which I am currently teaching. Upon concluding my postdoctoral appointment at SUSTech, I will formally join IIT Dharwad as a **Visiting Assistant Professor (VAP)**.

Ph.D. Thesis	<i>Risk-sensitive Stochastic Control and Games</i>
Primary Interests	Stochastic control, game theory, Markov decision processes, risk-sensitive control, singular perturbations, partially observable control, and stochastic differential games
Current Position	Postdoctoral Fellow, SUSTech, Shenzhen, China
Current Academic Engagement	Invited Course Instructor, IIT Dharwad
Forthcoming Position	Visiting Assistant Professor (VAP), IIT Dharwad
Languages	Bengali, English, and Hindi

Expertise and Skills

Research Expertise	Risk-sensitive stochastic control; zero-sum and nonzero-sum stochastic games; continuous-time and discrete-time Markov decision processes; piecewise deterministic Markov decision processes; singularly perturbed systems; partially observable stochastic control
Computational Methods	Monte Carlo simulation, Markov Chain Monte Carlo (MCMC), and numerical implementation of stochastic processes
Programming	Python and C
Academic Service	Reviewer for <i>Mathematical Methods of Operations Research</i> , <i>Systems & Control Letters</i> , <i>Stochastic Analysis and Applications</i> , and <i>International Journal of Game Theory</i>

Professional and Academic Experience

Postdoctoral Fellow

Dec 2024–Present

Southern University of Science and Technology (SUSTech), Shenzhen, China

Supervisor: Prof. Jie Xiong. Research on singularly perturbed stochastic control, partially observable stochastic control, and stochastic differential games.

Invited Course Instructor

Aug 2026–Present

Indian Institute of Technology Dharwad, India

Invited to serve as instructor for *Introduction to Mathematical Finance I* and *Stochastic Models*, which I am currently teaching.

Visiting Researcher <i>Indian Institute of Technology Bhubaneswar, India</i> Invited for collaborative research with Prof. Avijit Dutt Banik.	May 2026
Postdoctoral Fellow <i>Indian Institute of Technology Bombay, Mumbai, India</i> Supervisor: Prof. K. Suresh Kumar. Research on singular perturbations in stochastic control and games.	Jul 2023–Nov 2024
JRF Research Scholar <i>Indian Institute of Engineering Science and Technology, Shibpur, India</i> Research fellowship supported by CSIR.	Jul 2017–Jul 2018

Academic Qualifications

Ph.D. in Mathematics <i>Indian Institute of Technology Guwahati, India</i> Coursework CPI: 8.52; thesis: <i>Risk-sensitive Stochastic Control and Games</i> .	2023
M.Sc. in Applied Mathematics <i>Indian Institute of Engineering Science and Technology, Shibpur, India</i> 88%; CGPA: 9.3; special paper: Operations Research.	2016
B.Sc. in Mathematics (Honours) <i>Narasinha Dutt College, Howrah, India</i> 74.875%.	2014
Diploma in Information Technology Applications <i>Begri Youth Computer Training Centre (YCTC), Howrah, India</i> One-year diploma, January–December 2011.	2011
Higher Secondary Examination <i>Khasmora High School, Howrah, India</i> 83.4%.	2011
Secondary Examination <i>Rudrapur Siksha Sadan, Howrah, India</i> 78.62%.	2009

Teaching Experience

Course Instructor Stochastic Models <i>Indian Institute of Technology Dharwad, India</i> Invited Course Instructor; independently responsible for teaching and delivery of the course.	Aug 2026–Present
Introduction to Mathematical Finance I <i>Indian Institute of Technology Dharwad, India</i> Invited Course Instructor; independently responsible for teaching and delivery of the course.	Aug 2026–Present
Teaching Assistantships Probability Theory and Random Processes (MA225) <i>Indian Institute of Technology Guwahati</i> Undergraduate course.	Aug 2019; Aug 2021
Advanced Probability (MA684) <i>Indian Institute of Technology Guwahati</i> Ph.D. course.	Feb 2020
Probability Theory (MA590) <i>Indian Institute of Technology Guwahati</i> M.Sc. course.	Aug 2020
Discrete-Time Markov Chains and Poisson Processes <i>NPTEL / Indian Institute of Technology Guwahati</i> NPTEL online course.	Jan 2022
Probability I (SI427) <i>Indian Institute of Technology Bombay</i> M.Sc. course.	Aug 2023; Aug 2024

Fellowships, Qualifications and Honours

- SUSTech Postdoctoral Fellowship, supported by Shenzhen Municipality and Guangdong Province (2024–Present).
- IIT Bombay Institute Postdoctoral Fellowship (2023–2024).
- IIT Guwahati iPhD Fellowship, Ministry of Human Resource Development (2018–2023).
- NBHM Ph.D. Scholarship (2019–2020).
- CSIR Junior Research Fellowship, IEST Shibpur (2017–2018).
- Qualified CSIR-UGC NET in Mathematics five times, including four qualifications with JRF (2016–2018).
- GATE in Mathematics (2017).
- West Bengal Government Merit-cum-Means Scholarship during B.Sc. (2012–2014) and M.Sc. studies.

Selected Academic Engagements

- Seminar, IIT Bhubaneswar, India (2026) – delivered a talk.
- Mathematics Seminar, McGill University, Montreal, Canada (2024) – delivered a keynote.
- Workshop, Alfréd Rényi Institute of Mathematics, Budapest, Hungary, January 29–31, 2024 – delivered an invited talk.
- International Conference on Advances in Applied Probability and Stochastic Processes, St. Aloysius College, Thrissur, India, January 18–20, 2024 – presented a contributed talk.
- Probability and Stochastic Processes, ICTS Bangalore, India, September 17–21, 2023.
- Winter School on Graphs and Stochastic Processes, IISER Pune, India, December 16–20, 2019.
- Summer Outreach Program in Mathematics, NISER Bhubaneswar, India, June 20–July 2, 2016.

Research Collaboration

Collaborative work has been undertaken with Dr. Chandan Pal, Dr. Subhamay Saha, Prof. Mrinal K. Ghosh, Dr. Somnath Pradhan, Dr. Manikandan Rangaswamy, Dr. Abhay Sobhanan, and Prof. Jie Xiong.

Research Publications

Published and Accepted Articles

1. **Subrata Golui** and Chandan Pal. “Continuous-time zero-sum games for Markov decision processes with discounted risk-sensitive cost criterion on a general state space.” *Stochastic Analysis and Applications* (2021). DOI: [10.1080/07362994.2021.2013889](https://doi.org/10.1080/07362994.2021.2013889). Citation: 7; Q2.
2. **Subrata Golui** and Chandan Pal. “Continuous-time zero-sum games for Markov chains with risk-sensitive finite-horizon cost criterion.” *Stochastic Analysis and Applications* (2021). DOI: [10.1080/07362994.2021.1889381](https://doi.org/10.1080/07362994.2021.1889381). Citation: 12; Q2.
3. **Subrata Golui**, Chandan Pal, and Subhamay Saha. “Continuous-time zero-sum games for Markov decision processes with discounted risk-sensitive cost criterion.” *Dynamic Games and Applications* (2021). DOI: [10.1007/s13235-021-00391-2](https://doi.org/10.1007/s13235-021-00391-2). Citation: 8; Q2.
4. **Subrata Golui** and Chandan Pal. “Risk-sensitive discounted cost criterion for continuous-time Markov decision processes on a general state space.” *Mathematical Methods of Operations Research* (2022). DOI: [10.1007/s00186-022-00779-9](https://doi.org/10.1007/s00186-022-00779-9). Citation: 6; Q2.
5. Mrinal K. Ghosh, **Subrata Golui**, Chandan Pal, and Somnath Pradhan. “Nonzero-sum risk-sensitive continuous-time stochastic games with ergodic costs.” *Applied Mathematics and Optimization* (2022). DOI: [10.1007/s00245-022-09878-9](https://doi.org/10.1007/s00245-022-09878-9). Citation: 7; Q1.
6. Mrinal K. Ghosh, **Subrata Golui**, Chandan Pal, and Somnath Pradhan. “Discrete-time zero-sum games for Markov chains with risk-sensitive average cost criterion.” *Stochastic Processes and their Applications* (2022). DOI:

10.1016/j.spa.2022.12.009. Citation: 11; Q1.

7. Mrinal K. Ghosh, **Subrata Golui**, Chandan Pal, and Somnath Pradhan. "Zero-sum games for continuous-time Markov decision processes with risk-sensitive average cost criterion." *Mathematical Control and Related Fields* (2023). DOI: 10.3934/mcrf.2023003. Citation: 2; Q2.
8. **Subrata Golui**, Chandan Pal, Manikandan Rangaswamy, and Abhay Sobhanan. "Optimal control of a dynamic production-inventory system with various cost criteria." *Annals of Operations Research* (2023). DOI: 10.1007/s10479-023-05716-5. Citation: 3; Q1.
9. **Subrata Golui**. "Zero-sum games for piecewise deterministic Markov decision processes with risk-sensitive finite-horizon cost criterion." *Stochastic Analysis and Applications* (2025). DOI: 10.1080/07362994.2025.2470941. Citation: 0; Q2.
10. Mrinal K. Ghosh, **Subrata Golui**, Chandan Pal, and Somnath Pradhan. "An eigenvalue approach to risk-sensitive nonzero-sum ergodic games for discrete-time Markov decision processes." *Pure and Applied Functional Analysis* (2025). Accepted. Citation: 0; Q4.
11. **Subrata Golui** and Jie Xiong. "Continuous-time zero-sum games for Markov decision processes under the risk-sensitive first passage discounted cost criterion." *Dynamic Games and Applications* (2025). DOI: 10.1007/s13235-025-00677-9. Citation: 0; Q2.

Manuscripts Under Review

12. **Subrata Golui** and Jie Xiong. "Nonzero-sum risk-sensitive stochastic games for continuous-time Markov chains with discounted cost criterion." Under review.
13. **Subrata Golui**, Chandan Pal, Somnath Pradhan, and Jie Xiong. "Zero-sum games under the risk-sensitive discounted cost criterion for piecewise deterministic Markov decision processes." Under review.
14. **Subrata Golui** and Jie Xiong. "Discounted zero-sum games for singularly perturbed discrete-time Markov chains." Under review.

Personal Information and Academic References

Personal Information

Nationality	Indian
Date of Birth	21 June 1993
Parents	Gobordhan Golui and Radha Golui
Present Address	SUSTech, Nanshan District, Shenzhen, Guangdong, China – 518055
Permanent Address	Rudrapur, Domjur, Howrah, West Bengal, India – 711411

Academic References

Dr. Chandan Pal	Ph.D. Supervisor, IIT Guwahati
Prof. K. Suresh Kumar	Former Postdoctoral Mentor, IIT Bombay
Prof. Mrinal K. Ghosh	Collaborator, IISc Bengaluru
Dr. Subhamay Saha	Collaborator, IIT Guwahati